

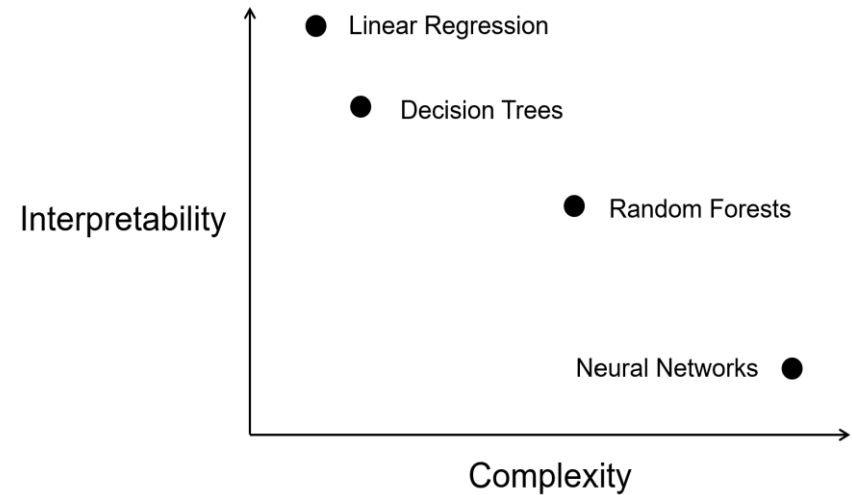
Attribution for Explainability of Black-Box ML Models

Ikhtiyor NEMATOV

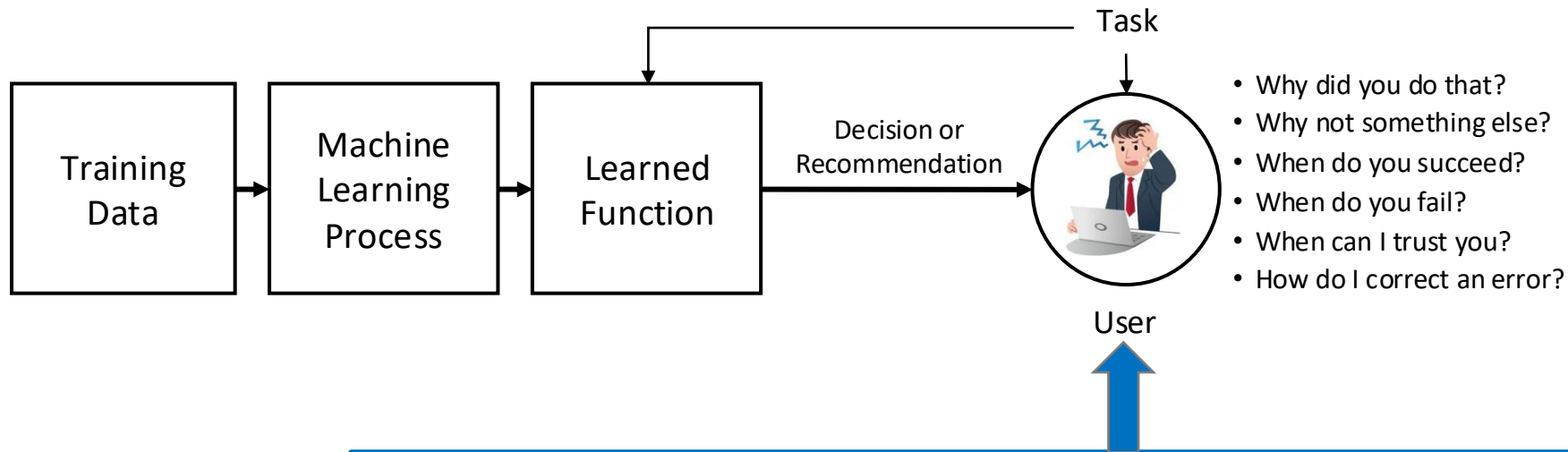
Elizaveta KUZMENKO

Why explainability?

- AI is everywhere
 - 📱 **Social media & search** – content recommendations and ranking
 - 🏦 **Finance** – loan approvals and fraud detection
 - 🏥 **Healthcare** – medical diagnosis support
 - 🚗 **Transportation** – navigation and autonomous driving
 - 🛍️ **Online services** – product and movie recommendations
- As models are tackling harder tasks, their complexity grows
- The more performant models are opaque



Explainable AI



- The target of XAI is an end user who:
 - depends on decisions, recommendations, or actions of the system
 - needs to understand the rationale for the system's decisions to understand, appropriately trust, and effectively manage the system
 - practitioner who needs to debug and inspect the model

Explainable AI

Human-understandable explanations of **outcomes** of algorithmic decision-making systems

Explanations that

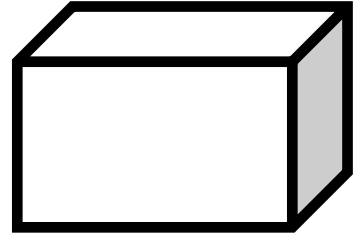
- enable understanding of overall strengths & weaknesses
- convey an understanding of how the system will behave in the future
- convey how to correct the system's mistakes

XAI is used to:

- detect biases and prevent discrimination
- detect training data errors, debug the model
- comply with regulations for transparency

White-Box vs Black-Box

- a **white box** is a transparent AI system
 - complete knowledge of inner workings
 - often interpretable by humans = interpretation
- a **black box** is an opaque, complex AI system
 - no knowledge of inner workings
 - we can only try to make an educated guess about how it reaches a decision = explanation

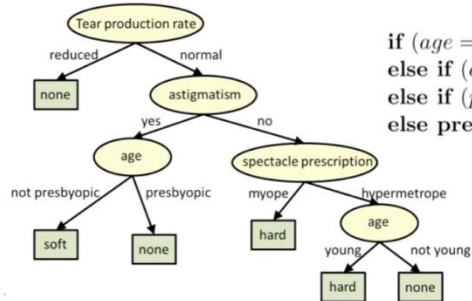


white box



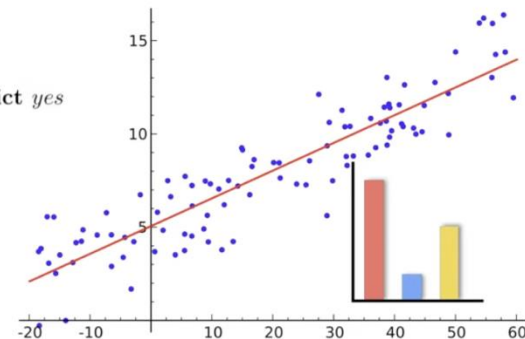
black box

How to achieve Explainability

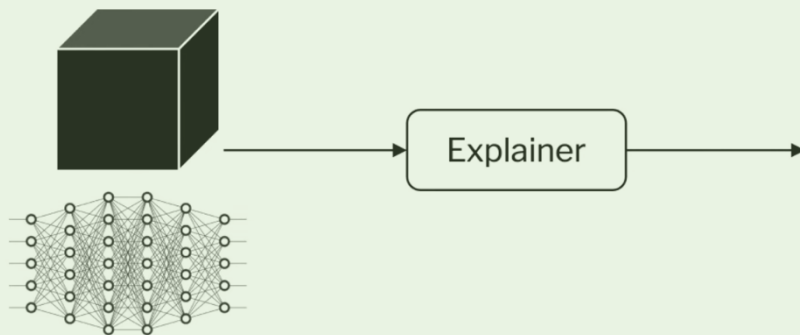


if (*age* = 18 – 20) and (*sex* = male) then predict *yes*
 else if (*age* = 21 – 23) and (*priors* = 2 – 3) then predict *yes*
 else if (*priors* > 3) then predict *yes*
 else predict *no*

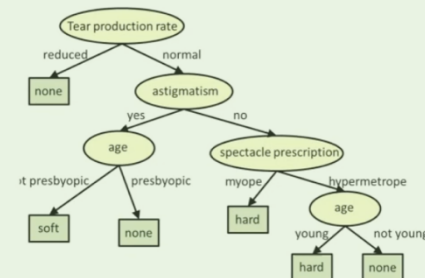
Intrinsic



Post hoc



if (*age* = 18 – 20) and (*sex* = male) then predict *yes*
 else if (*age* = 21 – 23) and (*priors* = 2 – 3) then predict *yes*
 else if (*priors* > 3) then predict *yes*
 else predict *no*



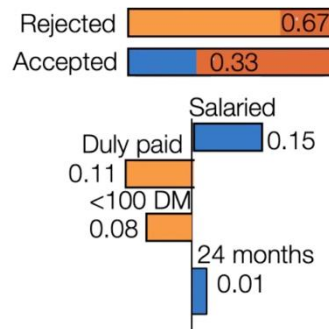
Types of Explanations

Name	Age	Saving	Month	...	Credit History
Maeve	<25	<100 DM	24	...	Paid duly

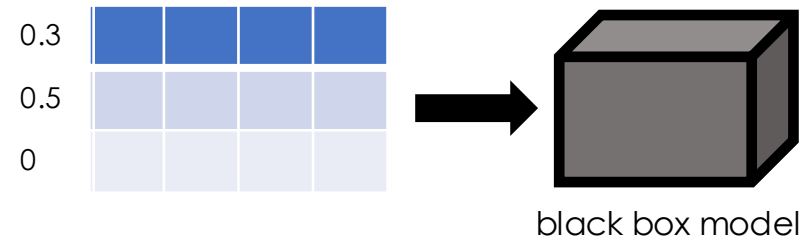


Why was my loan application rejected?

Feature attribution Explanations

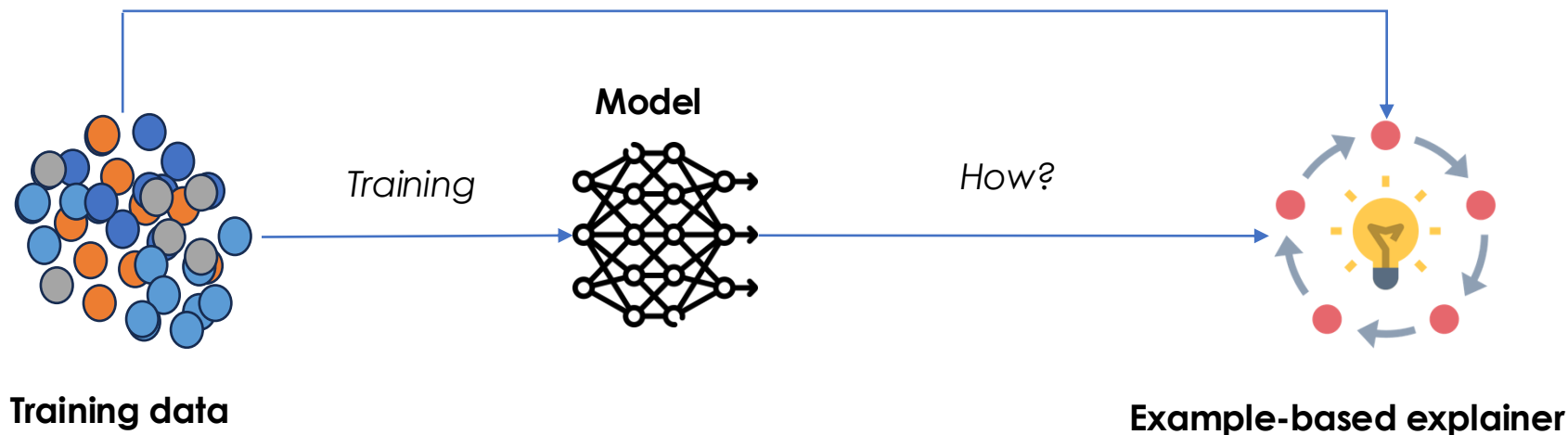


Data attribution Explanations



Data Attribution Explainability

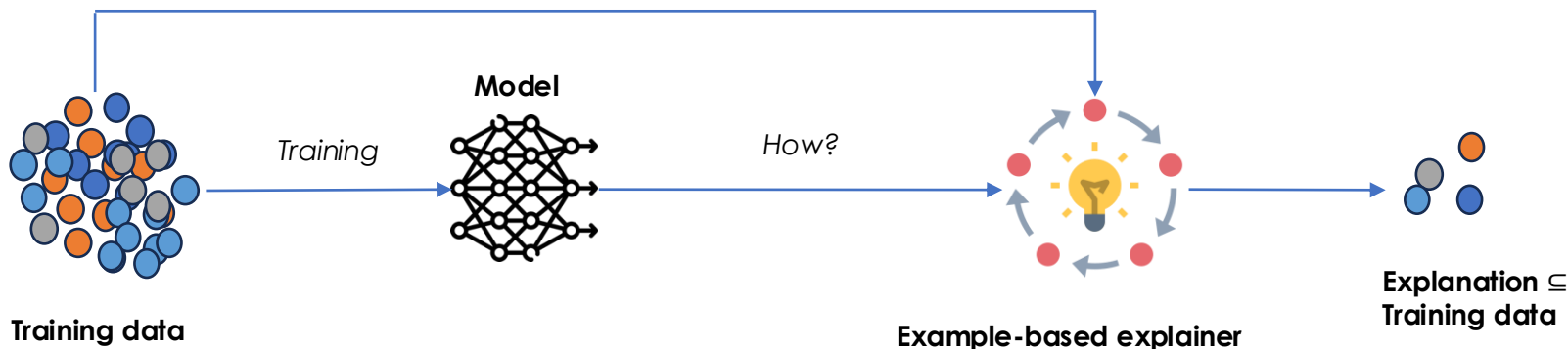
- Explaining the model through the lens of the **data** it has been trained on
- Can be **local**, explains a specific prediction, or **global** explains models behavior



Data Attribution Explainability

Explaining the model through the lens of the **data** it has been trained on

Can be **local**, explains a specific prediction, or **global** explains models behavior



Global Data Attribution Explainability

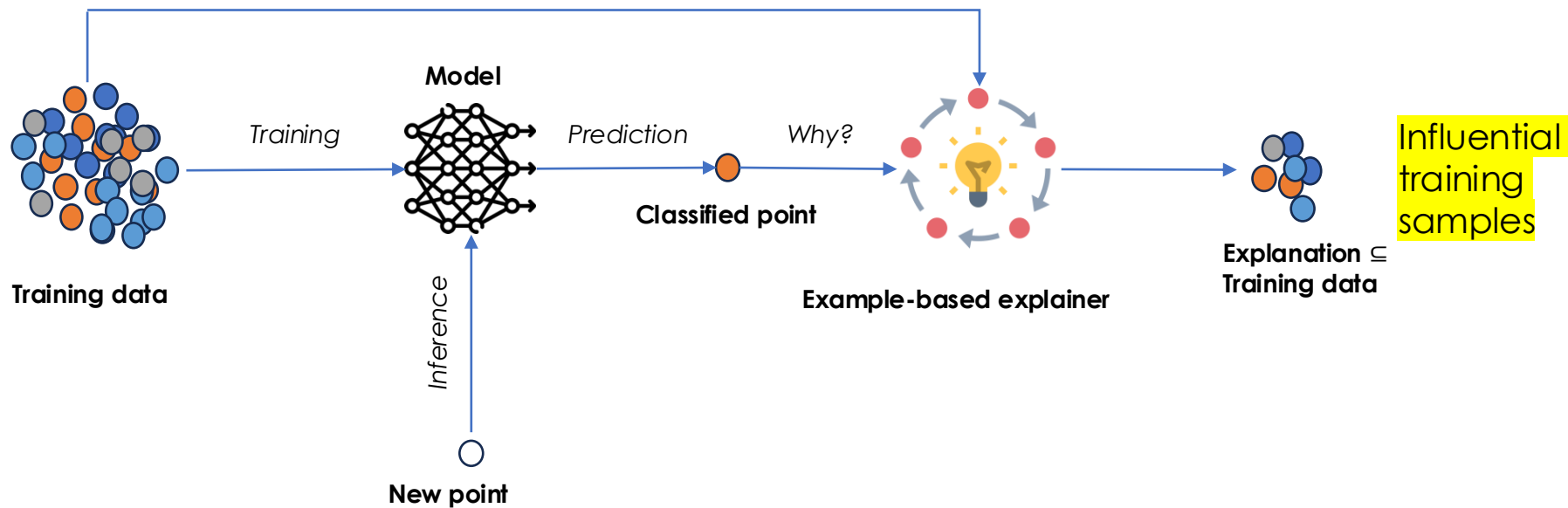


**Explanation $\subseteq$
Training data**

Prototypes, Representative points

Explaining by affinity to this prototypes

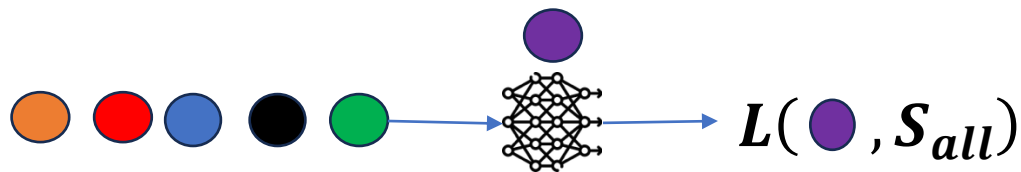
Local Data Attribution Explainability



Local Data Attribution Explainability Methods

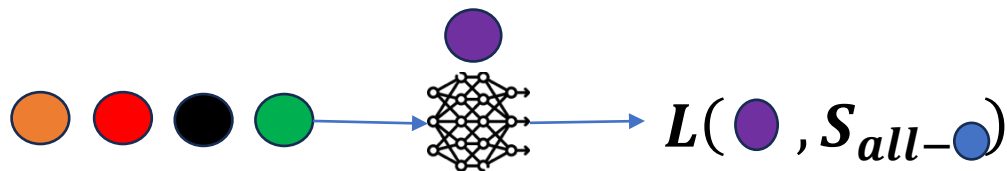
Leave-One-Out

- Remove and retrain
- Change in the model performance w.r.t. the prediction of interest
 - Loss : $L(S_{all}) - L(S_{all} - i^{th})$
- For each prediction remove and retrain each training sample
- Exhaustively expensive for large datasets



N

What is the influence of ● on the prediction for ● ? $\Rightarrow L(\text{purple circle}, S_{all-\text{blue circle}}) - L(\text{purple circle}, S_{all})$



Influence Function

- Given data points $z \in Z$ and loss function L , the learning algorithm searches for a model with parameters $\theta \in \Theta$ such that

$$\theta^* = \operatorname{argmin}_{\theta \in \Theta} \frac{1}{n} \sum_{i=1}^n L(z_i, \theta)$$

- Influence of training data point z_{train} (from robust statistics)
 - How do the model parameters change as we upweight z_{train} by an infinitesimal amount ϵ ?

$$\theta_\epsilon = \operatorname{argmin}_{\theta \in \Theta} \frac{1}{n} \sum_{i=1}^n L(z_i, \theta) + \epsilon L(z_{\text{train}}, \theta)$$

Influence Function

- Original model parameters: θ , updated model parameters: θ_ϵ
- Influence of z_{train} on test loss is then measured as

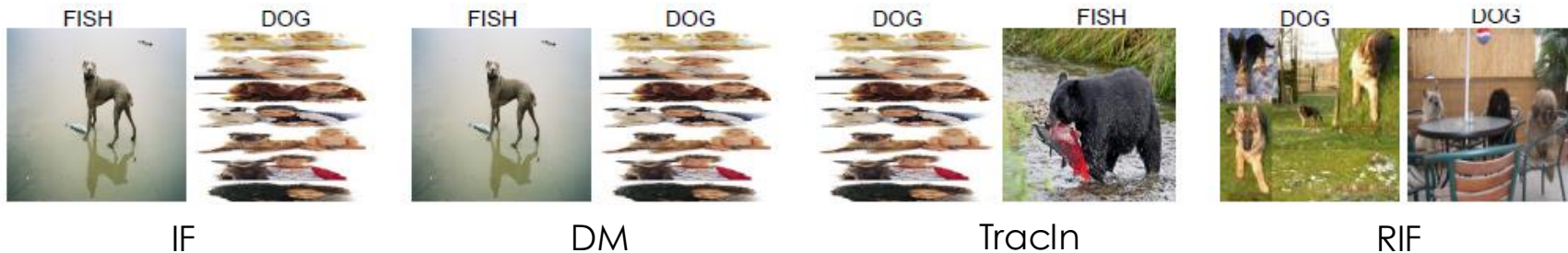
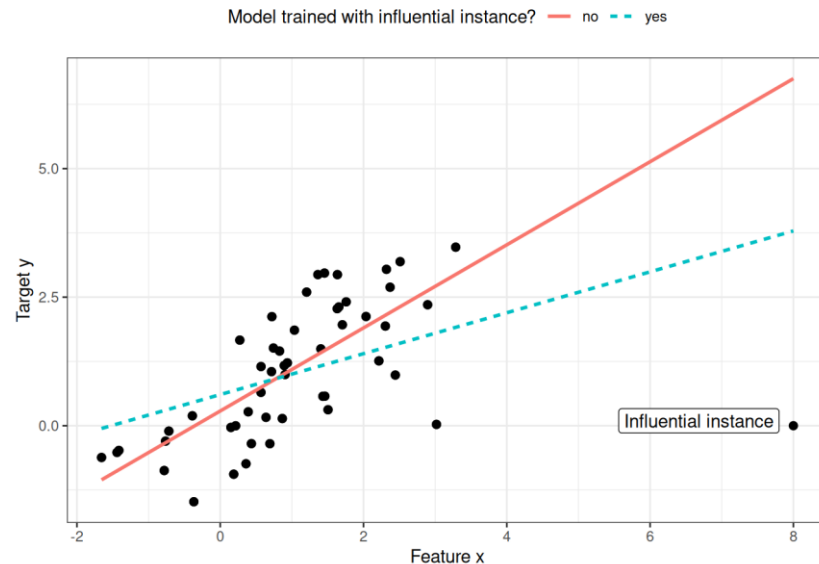
$$\mathcal{I}_{\text{up,loss}}(z_{\text{train}}, z_{\text{test}}) = L(z_{\text{test}}, \theta_\epsilon) - L(z_{\text{test}}, \theta^*)$$

$$\begin{aligned} &\stackrel{\text{def}}{=} \left. \frac{dL(z_{\text{test}}, \hat{\theta}_{\epsilon, z_{\text{train}}})}{d\epsilon} \right|_{\epsilon=0} \\ &= - \underbrace{\nabla_{\theta} L(z_{\text{test}}, \hat{\theta})^\top}_{\text{Gradient of test loss}} H_{\hat{\theta}}^{-1} \underbrace{\nabla_{\theta} L(z_{\text{train}}, \hat{\theta})}_{\text{Gradient of training data loss}}, \end{aligned}$$

$$\text{where } H_{\hat{\theta}} \stackrel{\text{def}}{=} \underbrace{\frac{1}{n} \sum_{i=1}^n \nabla_{\theta}^2 L(z_i, \hat{\theta})}_{\text{Hessian matrix}}.$$

Robustness of Leave-One-Out Methods

- Since the explanation is subset of data it is impacted by the **data quality** e.g. **class-outliers**
- Class outliers are point that resemble one class but labelled as another, or exhibit affinity to both classes
- Such points are hard for the model to classify, thus have high loss



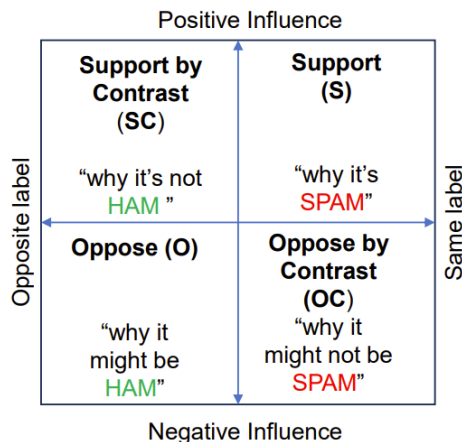
User Intent in Explainability

- RIF is suboptimal
- Proximity-based robustness
- Importance of negative influence
- Explaining by contrast
- Customizing according user intent

$$\mathcal{E}_q = \arg \max_{\mathcal{E} \subseteq \mathcal{S}_q, |\mathcal{E}|=k} \sum_{z \in \mathcal{E}} (\alpha |I(z, z_t)| + \beta P(z, z_t)) + \gamma D(\mathcal{E})$$

Explanation Intents:

- Interpreting a correct prediction => S, SC
- Investigating a wrong prediction => S, SC, O, OC;
- Clarifying an ambiguous prediction => S, O



Test Sample



Prediction : **Dog**

Ground Truth: **Fish**

AIDE



Shapley Values – Game Theory

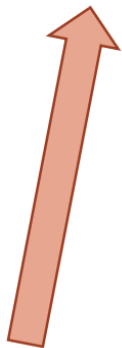
Shapley Values is the **unique** solution to the **fair allocation** of payoffs among members in a **coalition** participating in a game characterized by a specific set of axioms.



Lloyd Shapley
2012 Nobel Memorial Prize
in Economic Sciences

Shapley Values – Formula

$$\phi_i(v) = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(|N| - |S| - 1)!}{|N|!} (v(S \cup \{i\}) - v(S))$$



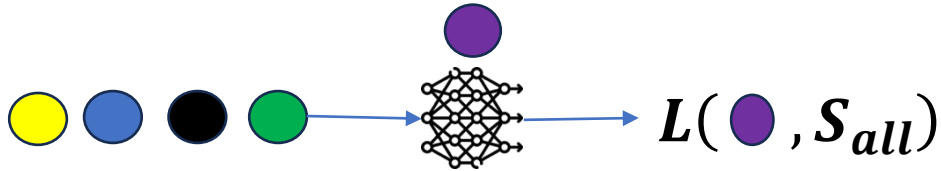
Shapley Value for
feature i

Iterating over all
possible
coalitions without
feature i

Weight of
coalition

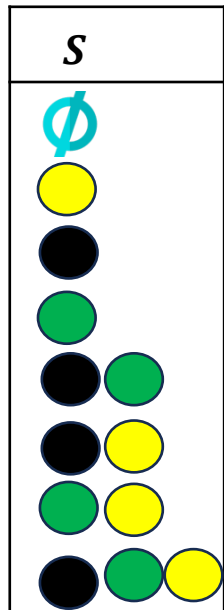
Marginal contribution of
feature i to this coalition
S

Shapley Values – Example



What is the influence of ● on the prediction for ●?

2^N



Shapley Value for feature i

$$\phi_i(v) = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(|N| - |S| - 1)!}{|N|!} (v(S \cup \{i\}) - v(S))$$

Iterating over all possible coalitions without feature i

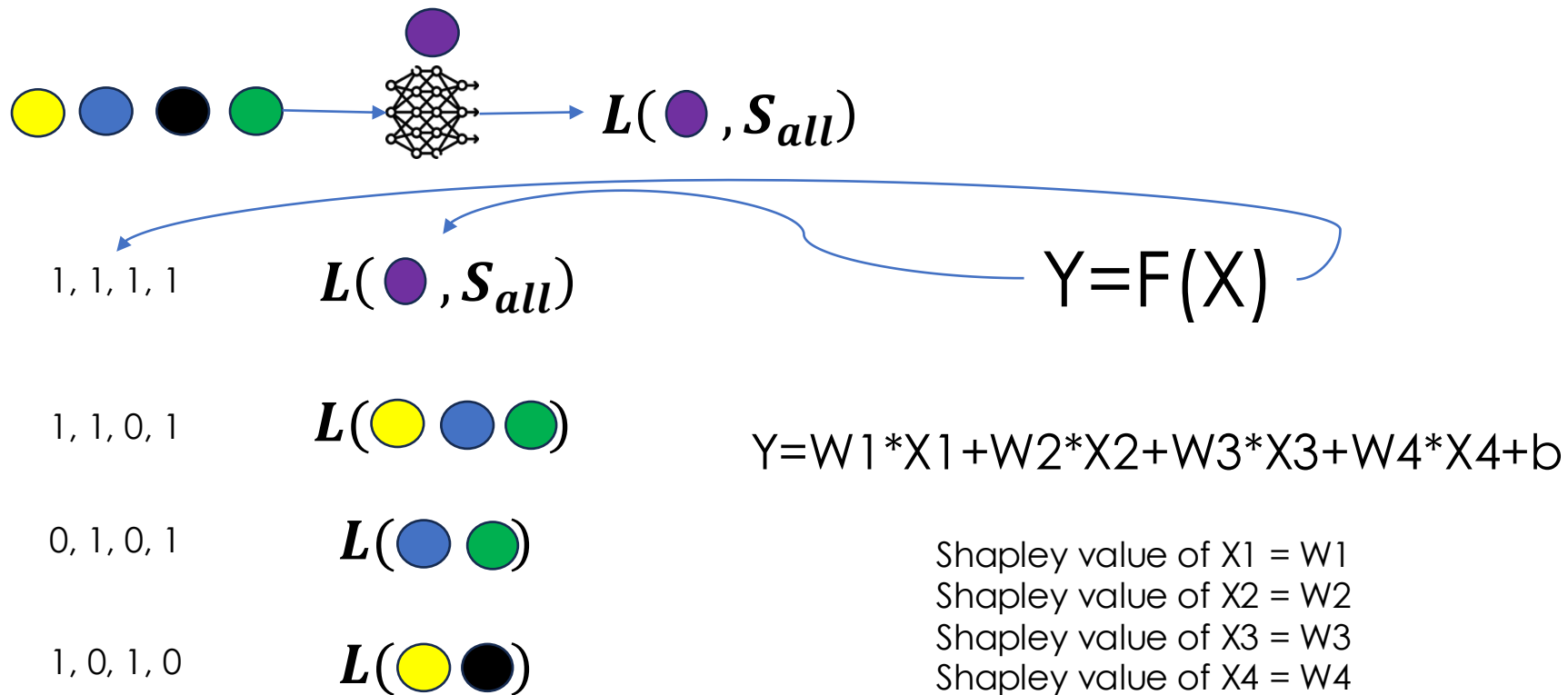
Weight of coalition

Marginal contribution of feature i to this coalition S

$$L(\bullet, S) - L(\bullet, S + \bullet)$$

Kernel-Shap

- Surrogate model for predicting the Loss



Shapley Values for Data – Efficiency

- Monte-Carlo method (Ghorbani et al.)

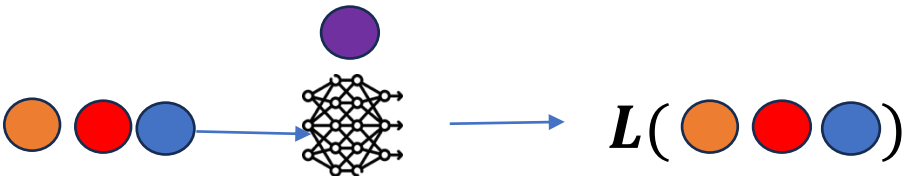
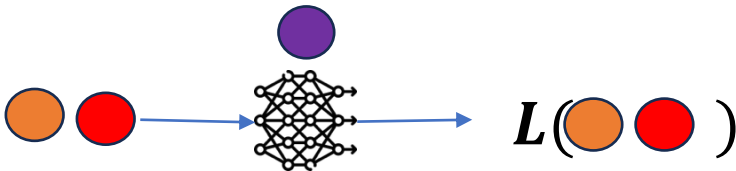
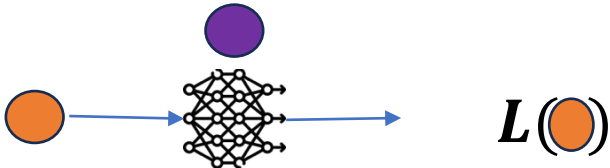
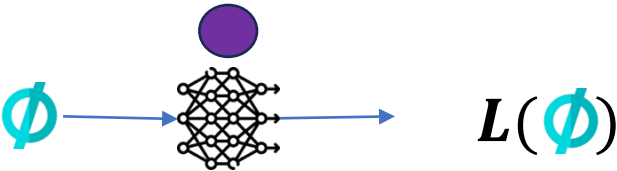
$$\text{Value } (t_i) = \mathbb{E}_{\pi \sim \Pi} [V(\underbrace{S_{\pi}^i}_{\text{Random permutation sampled from all permutations of data points}} \cup i) - V(\underbrace{S_{\pi}^i}_{\text{Data appearing before } i \text{ in a permutation}})]$$

Random permutation
sampled from
all permutations of
data points

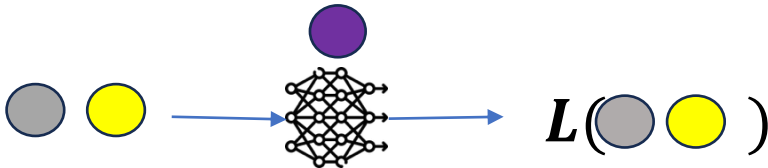
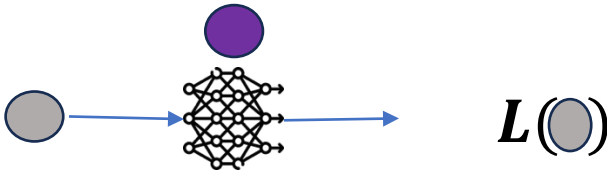
Data appearing
before i in a
permutation

- Truncated Monte Carlo Shapley: Use early stopping when calculating data Shapley for a sampled permutation

Example TMC-Shapley



.....



.....

$$\text{Value } (t_i) = \mathbb{E}_{\pi \sim \Pi}[V(S_{\pi}^i \cup i) - V(S_{\pi}^i)]$$

Global Data Attribution Explainability Methods

Data Prototypes

- Prototypes: data points that are representative of all the data



- Criticisms: data points not represented by prototypes



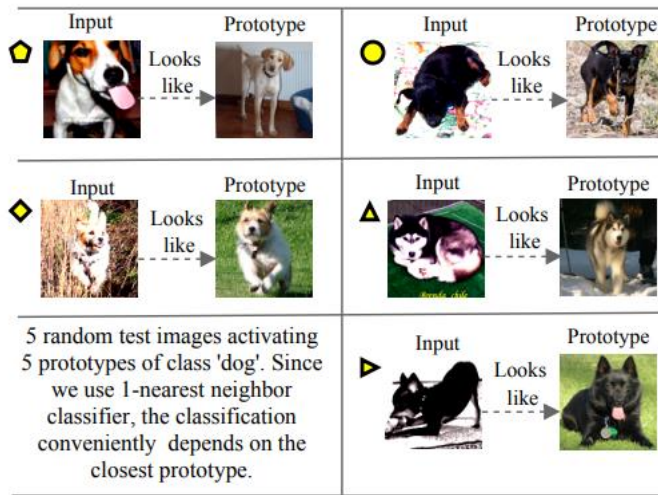
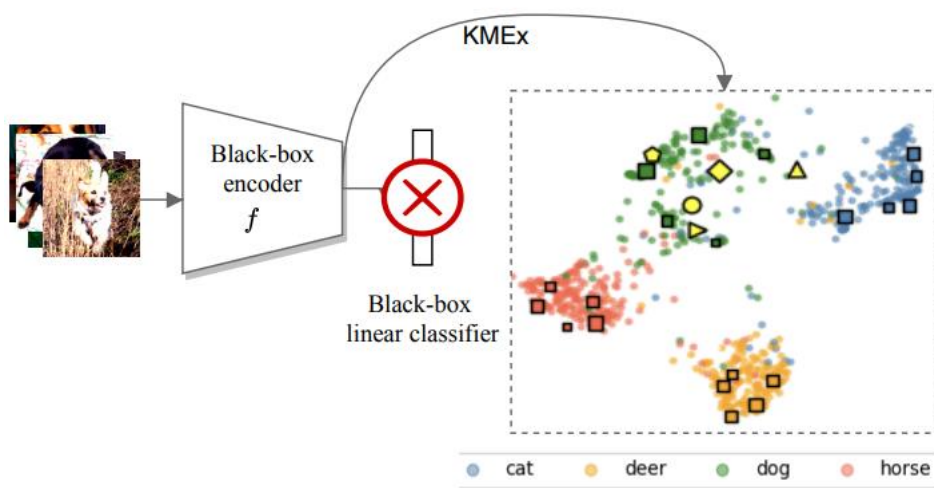
Representer Points

- Representer points
 - Decompose a predictor into a linear combination of functions of training data points
 - Weighted similarities between test data point and each training data point

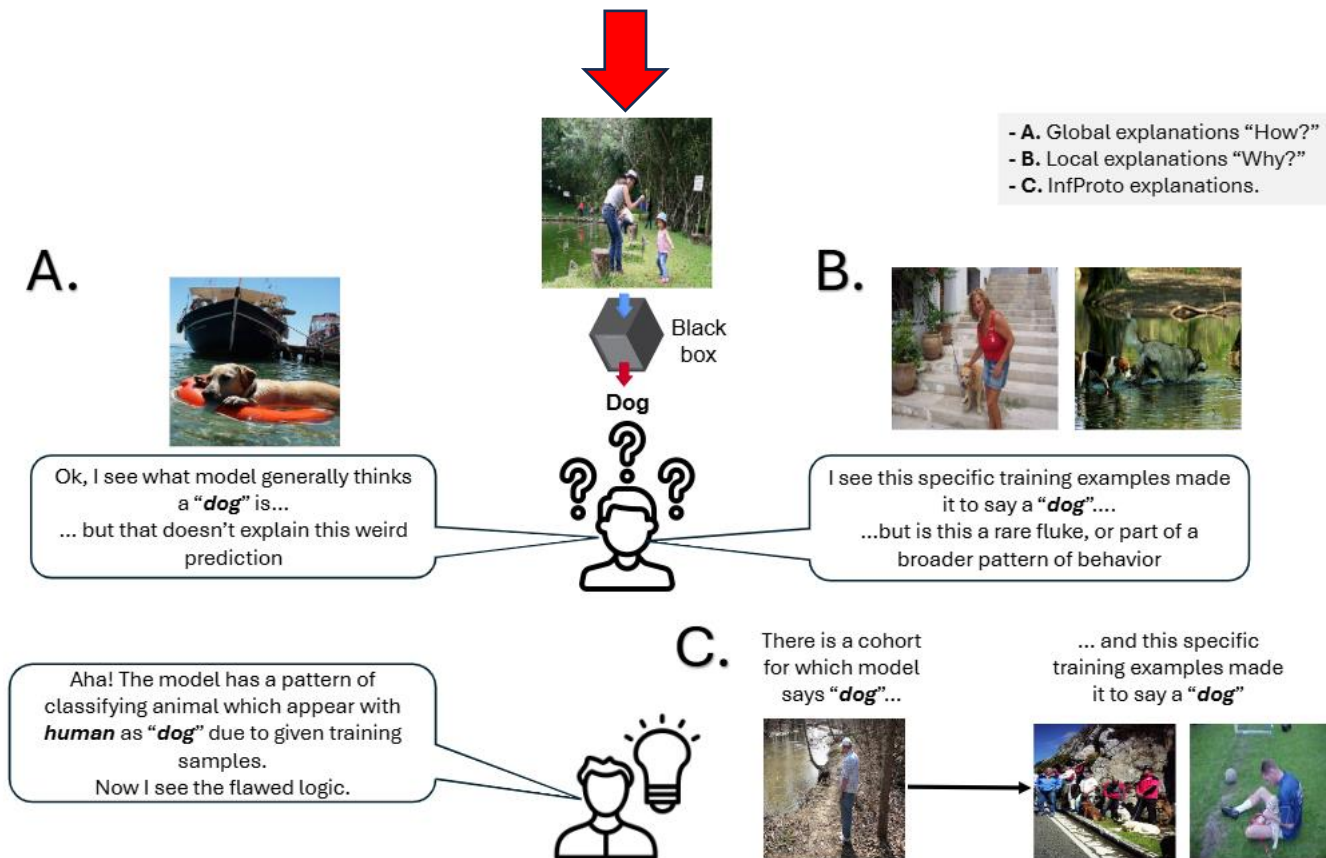


Prototypical Self-Explainable Models

- Finding prototypes from models embedding space
- Using the prototypes for classification
- High faithfulness to the model, high accuracy to the data



Bridging the Local and Global Explainability



Bridging the Local and Global Explainability

- one process generate both type
- robust version of influence function
- preferred by **68 %** of users

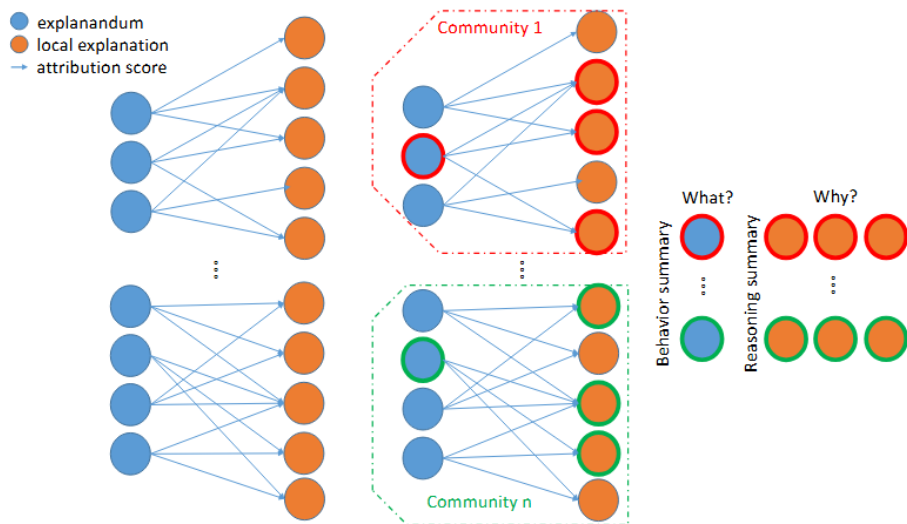
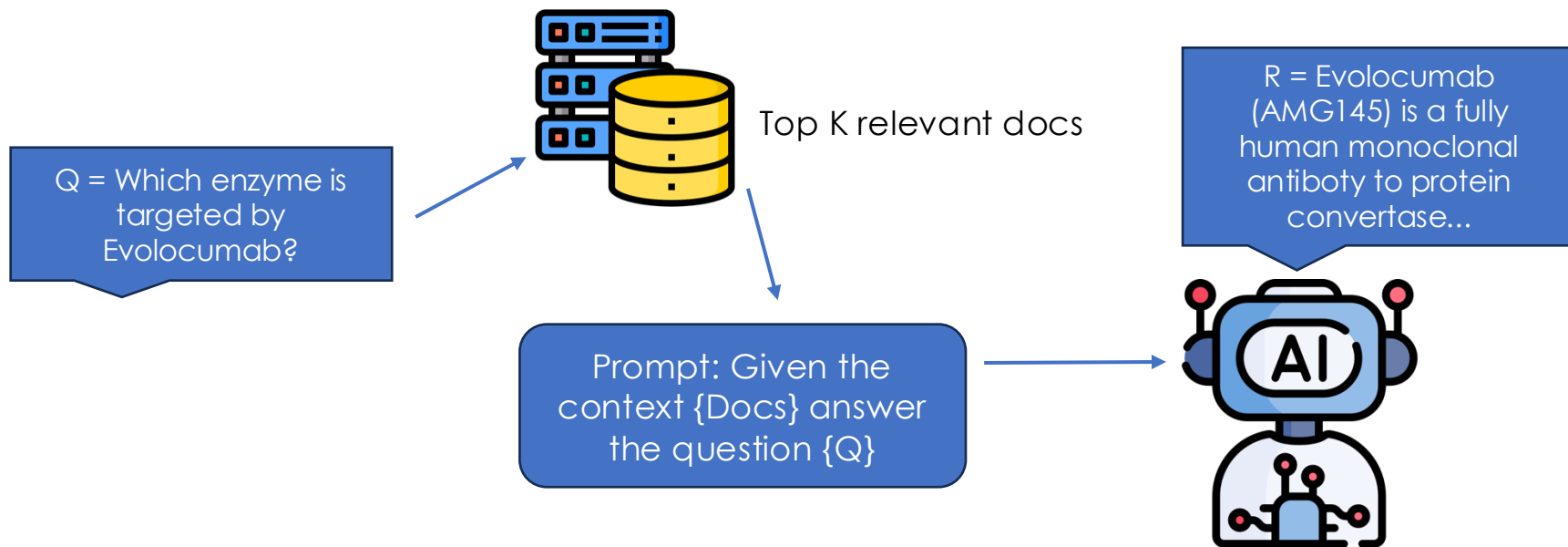


Fig : Schematic overview of the InfProto explanation generation process.

Explainability in the era of LLMs

RAG & LLMs

- Retrieval Augmented Generation (RAG) grounds LLMs in external knowledge, reducing hallucinations and improving factuality.

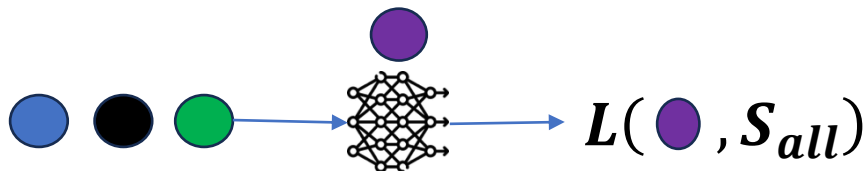


But which documents (if any) have actually influenced the final answer?

Attribution for LLMs

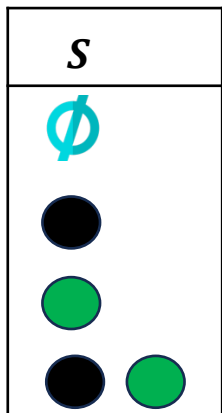
- **Gradient-based methods:** analyze the gradients of the output with respect to the input
- **Attention-based methods:** use attention scores within the transformer layers
- **Prompting-based methods:** ask an LLM to self-rationalize
- **Perturbation-based methods:** remove a doc and see what changes

Attribution for RAG: Shapley values



What is the influence of document ● on the prediction for ● ?

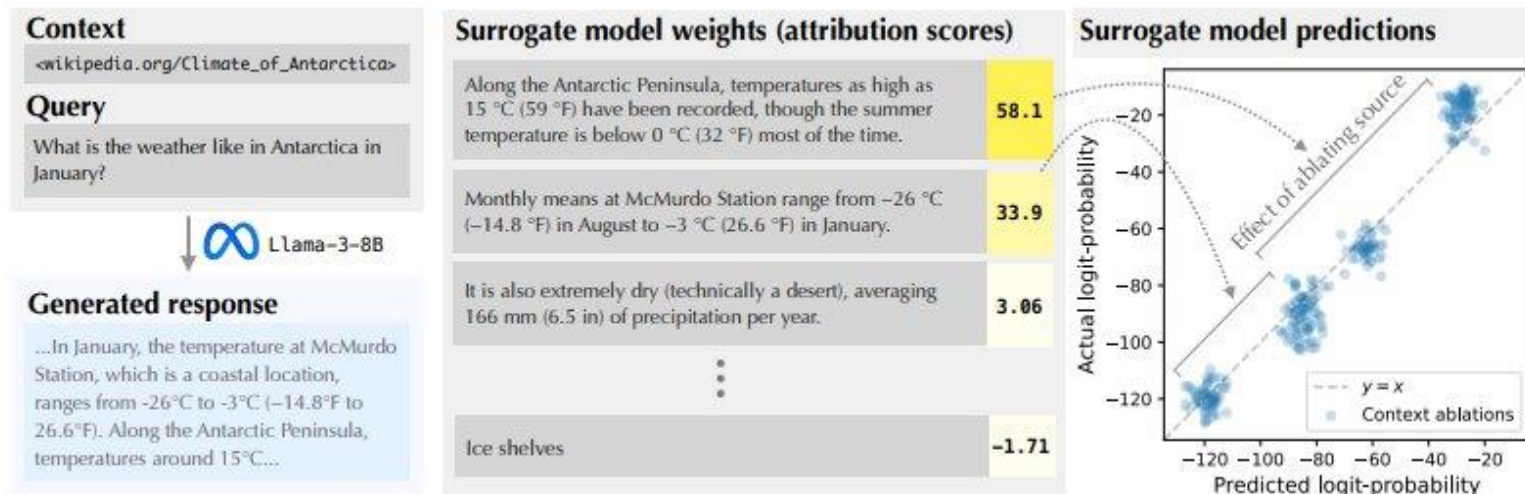
$$L(\text{●}, S + \text{●}) - L(\text{●}, S)$$



2^N LLM calls :(
Need to approximate

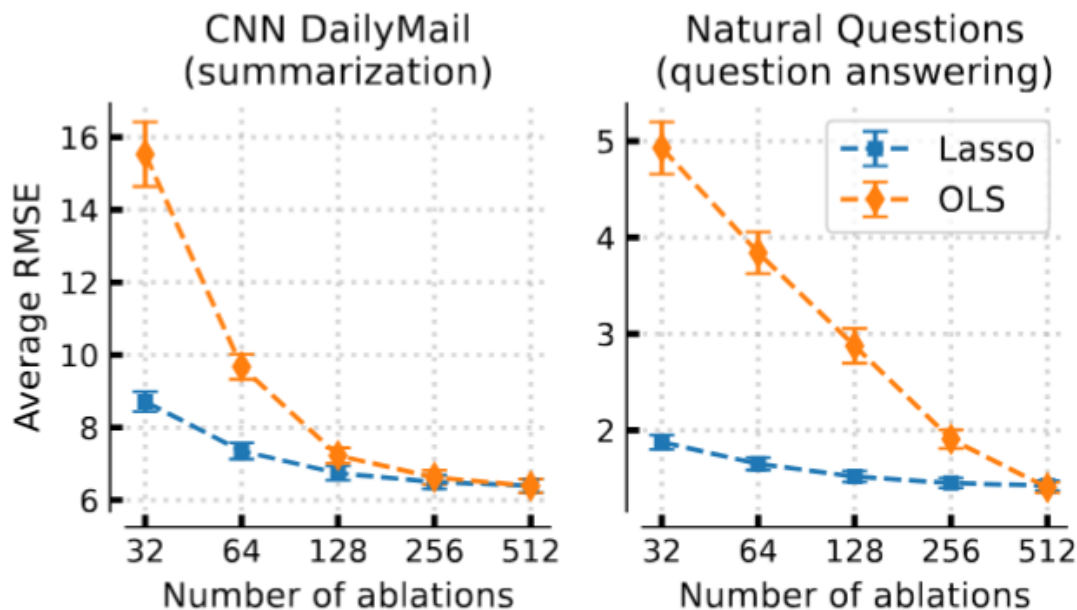
ContextCite

- Main idea: train a surrogate (linear) model that approximates how a LM's response is affected by including or excluding each part of the context.
- The weights of the surrogate model are treated as attribution scores.
- Focus on sentence level attribution, but the method can do any level (document, paragraph, sentence, word).



ContextCite

- Good performance for marginal attribution
- Does not account for interactions between documents



Inter-document relationships:

- **Redundancy:** What is the weather in Suvsambil?

1. *The weather in Suvsambil is sunny*
2. *Suvsambil is a mountainous country*
3. *The sun is shining in Suvsambil today*

- **Complementarity:** What are the roles or professions of Elara Vayne and Jax Korden?

1. *Elara Vayne is the chief Star-Navigator of the starship 'Wanderer'*
2. *Many young Squibs dream of joining the Sky Guard.*
3. *Jax Korden serves as the primary Rift-Warden protecting the Chronos Gate.*

- **Synergy (Multi-Hop):** What is the weather in the capital of Suvsambil?

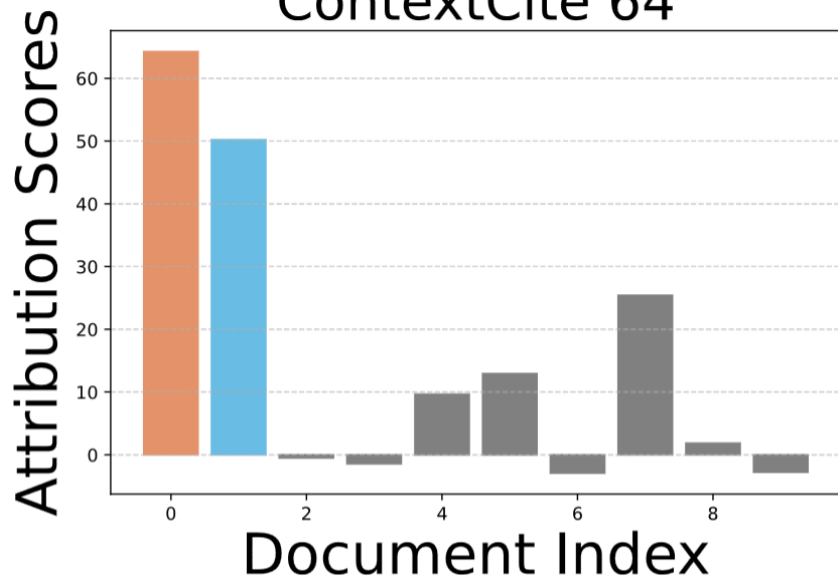
1. *The capital of Suvsambil is Savrak.*
2. *Weather in Tentak is cloudy.*
3. *Weather in Savrak is sunny.*

Source attribution & document interactions

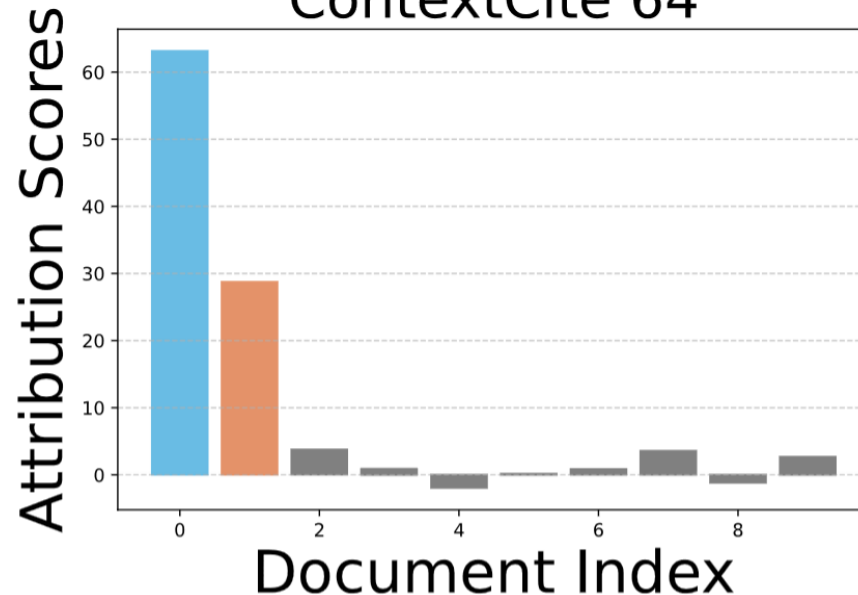
Redundancy

Synergy

ContextCite 64

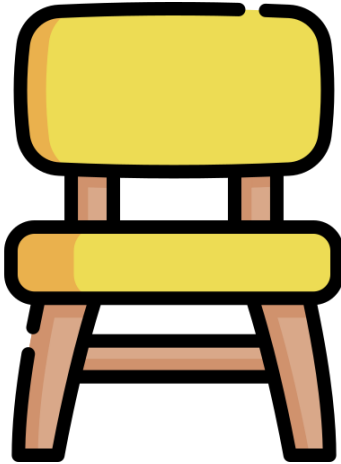


ContextCite 64



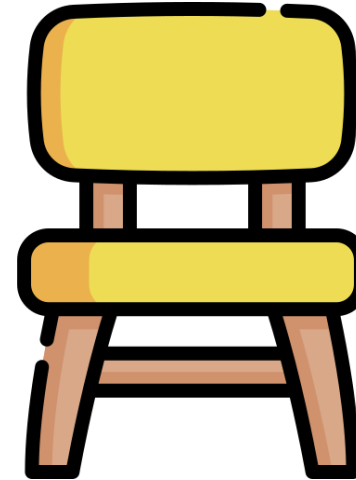
Source attribution & document interactions

Shapiq, Spex, ...



Modeling Efficiency

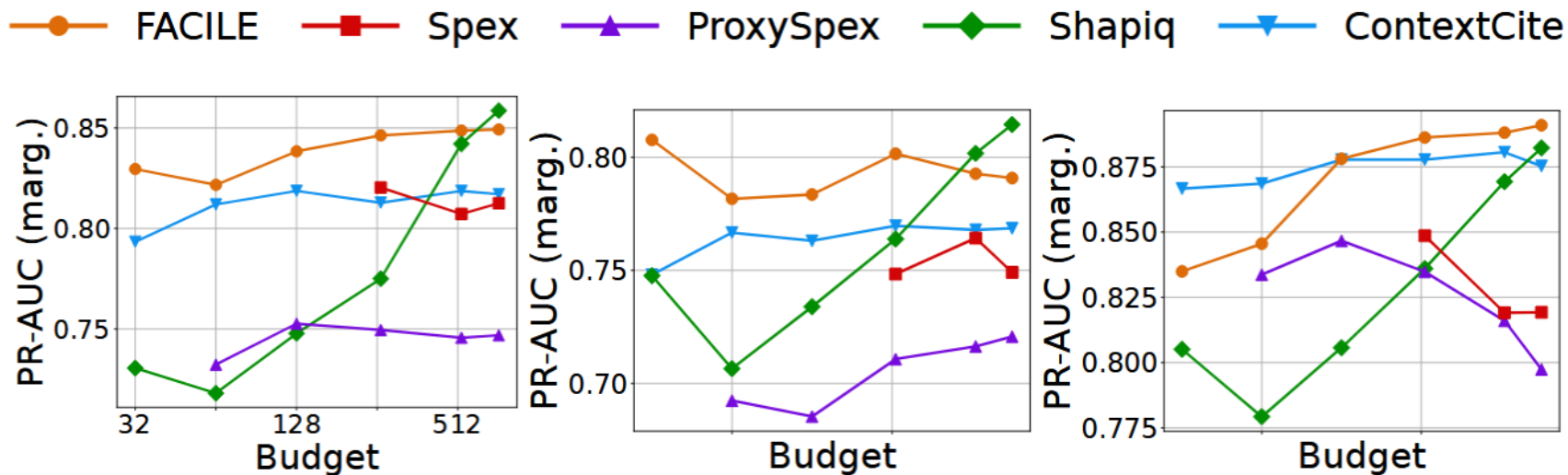
???



Cost Optimization

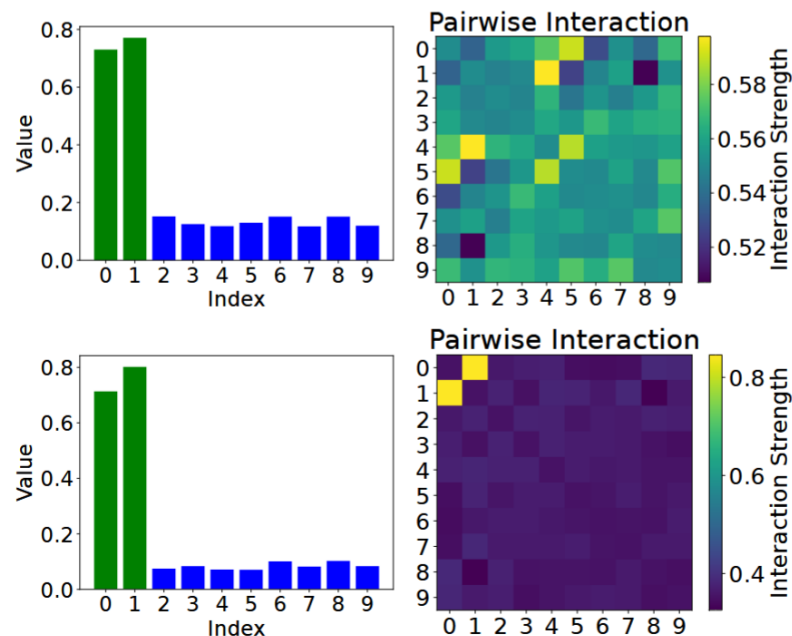
FACILE

- From linear model to polynomial: Factorization Machines as a surrogate model
- Maximum interaction order to two -> pairwise interactions



FACILE: Measuring Interactions

- Two documents receive high marginal attributions
- Interaction matrices distinguish the cases
- **Complementarity**: negligible interaction
- **Synergy**: strongest interaction among all document pairs



Questions?